

FARM Kerberos NFS 설정 runbook

Status: canonical FARM document Last reviewed: 2026-07-18 KST Runtime baseline last verified:
FARM mounts 2026-07-06, NAS keytab health 2026-07-18

This is the single current FARM runbook. It consolidates Samba AD DC, Synology NAS, FARM host setup, NFS mount, identity mapping, container provisioning, applied-state checks, and recovery guidance. Superseded FARM notes remain available in Git history; they do not override this file.

Before a state-changing operation, re-run the verification commands in this document. A "last verified" date records an observation, not a guarantee that the remote systems have not changed.

Current Applied Baseline

The current desired and last verified production data path is:

```
nas.farm.decs.internal:/volume1/share  
-> /home/takoX/share  
nfs4 vers=4.0 sec=krb5 proto=tcp hard timeo=600 retrans=2  
addr=100.100.100.120
```

FARM1/2/6/7/8/9 were verified with this runtime profile on 2026-07-06. The read-only NAS keytab checker reported healthy on 2026-07-18 02:02 KST: AD KVNO 4, both NAS keytabs current, the AILAB acceptor preserved, current ticket decryption successful, and `svcgssd` running without a `-p` restriction. Monitoring is read-only; repair and rotation remain explicit commands under `script/keytab/`.

Known residual hardening items remain: management-network NAS exports are looser than the production path, and AD replication health must be checked before relying on writes through a secondary DC. These are not reasons to use the management IP for production NFS.

0. Target Architecture

Samba AD DCs:

```
farm2 100.100.100.102
farm6 100.100.100.106
farm7 100.100.100.107
```

Synology NAS:

```
management SSH: 192.168.2.30:6954
NFS storage IP: 100.100.100.120
NFS source:      nas.farm.decs.internal:/volume1/share
user homes:      /volume1/share/user-share/<username>
```

FARM hosts:

```
farm1 /home/tako1/share clientaddr 100.100.100.101
farm2 /home/tako2/share clientaddr 100.100.100.102
farm6 /home/tako6/share clientaddr 100.100.100.106
farm7 /home/tako7/share clientaddr 100.100.100.107
farm8 /home/tako8/share clientaddr 100.100.100.108
farm9 /home/tako9/share clientaddr 100.100.100.109
```

```
Kerberos realm:      FARM.DECS.INTERNAL
DNS domain:          farm.decs.internal
NetBIOS domain:      FARM
NFS server principal: nfs/nas.farm.decs.internal@FARM.DECS.INTERNAL
Current NFS version: NFSv4.0
NFS security:        sec=krb5
```

Do not use `192.168.2.30` as the NFS transport address. That address is only for NAS management SSH. FARM NFS mounts must resolve and route `nas.farm.decs.internal` to `100.100.100.120`.

2026-06-29 Rollout Verification

Checked with Ansible against the FARM inventory:

```
farm1 /home/tako1/share -> nas.farm.decs.internal:/volume1/share
farm2 /home/tako2/share -> nas.farm.decs.internal:/volume1/share
farm6 /home/tako6/share -> nas.farm.decs.internal:/volume1/share
farm7 /home/tako7/share -> nas.farm.decs.internal:/volume1/share
farm8 /home/tako8/share -> nas.farm.decs.internal:/volume1/share
farm9 /home/tako9/share -> nas.farm.decs.internal:/volume1/share
```

The expected FARM mount identity is:

```
fstype=nfs4
vers=4.0
sec=krb5
addr=100.100.100.120
clientaddr=100.100.100.10x
```

As of 2026-07-06 10:25 KST, farm1/2/6/7/8/9 had the desired `sec=krb5,timeo=600,retrans=2` runtime profile.

At that verification, all FARM hosts had AD/NFS identity lookup wired through NSS:

```
passwd: files systemd winbind
group:  files systemd winbind
rpc-gssd: active
```

Observed service split in that verification:

```
farm2/farm6/farm7: samba-ad-dc active, standalone winbind inactive
farm1/farm8/farm9: samba-ad-dc inactive, standalone winbind active
```

This matches the intended model: AD DC hosts use `samba-ad-dc`; non-DC FARM clients use standalone `winbind` for NSS identity lookup.

1. Required Identity Invariant

For every FARM Kerberos container, the same Unix identity must be used everywhere:

```
DB uid/gid
= AD uidNumber/gidNumber
= NAS RFC2307 uid/gid view
= FARM NFS returned uid/gid
= container uid/gid
```

Example:

```
username: zio_ascp
group:    ASCP

DB:       10073:10070
AD user:   FARM\zio_ascp uidNumber=10073 gidNumber=10070
AD group:  FARM\ASCP gidNumber=10070
NAS stat:  10073:10070
host stat: 10073:10070
container: uid=10073 gid=10070
```

Do not use Synology/Samba internal IDs such as `96470xxx` as DECS user IDs. If those values appear, RFC2307 mapping is not correctly active for that path or host.

2. Samba AD DC Setup

2.1 Primary DC

The current primary DC is farm2. The original PoC provisioning playbook has been removed from the active tree; use the commands and required values in this runbook as the current reference.

Important values:

```
ad_realm: FARM.DECS.INTERNAL
ad_dns_domain: farm.decs.internal
ad_netbios_domain: FARM
ad_dc_fqdn: dc1.farm.decs.internal
ad_dc_service_ip: 100.100.100.102
nas_fqdn: nas.farm.decs.internal
nas_storage_ip: 100.100.100.120
```

The primary DC must be provisioned with RFC2307 support:

```
sudo samba-tool domain provision \
  --server-role=dc \
  --use-rfc2307 \
  --dns-backend=SAMBA_INTERNAL \
  --realm=FARM.DECS.INTERNAL \
  --domain=FARM
```

Required packages:

```
sudo apt-get update
sudo apt-get install -y acl attr dnsutils krb5-user samba winbind
```

The AD DC service should be:

```
sudo systemctl enable --now samba-ad-dc
sudo systemctl disable --now smbd nmbd winbind 2>/dev/null || true
```

On an AD DC, identity lookup is served by `samba-ad-dc`; standalone `winbind.service` may be stopped or failed and that can be normal for DC mode.

2.2 Additional DCs

The original additional-DC PoC playbook has been removed from the active tree. Promote a new supported playbook only if additional DC provisioning becomes a regular operation.

Current additional DCs:

```
farm6 100.100.100.106
farm7 100.100.100.107
```

Join shape:

```
sudo samba-tool domain join FARM.DECS.INTERNAL DC \
-U 'FARM\Administrator' \
--realm=FARM.DECS.INTERNAL \
--dns-backend=SAMBA_INTERNAL \
--server=dc1.farm.decs.internal
```

Every DC must have `/etc/krb5.conf` listing all known KDCs:

```
[libdefaults]
    default_realm = FARM.DECS.INTERNAL
    dns_lookup_realm = false
    dns_lookup_kdc = false
    rdns = false
    ticket_lifetime = 24h
    renew_lifetime = 7d
    forwardable = true

[realms]
    FARM.DECS.INTERNAL = {
        kdc = 100.100.100.102
        kdc = 100.100.100.106
        kdc = 100.100.100.107
        admin_server = 100.100.100.102
        default_domain = farm.decs.internal
    }

[domain_realm]
    .farm.decs.internal = FARM.DECS.INTERNAL
    farm.decs.internal = FARM.DECS.INTERNAL
```

Replication should be checked before relying on farm6/farm7 for AD writes:

```
sudo samba-tool drs showrepl
```

During the current rollout, AD writes should be performed against farm2 until replication health is repaired and confirmed.

2.3 AD User And Group Policy

For a user with no separate shared group:

```
AD user:          FARM\<username>
AD private group: FARM\<username>_gid
principal:        <username>@FARM.DECS.INTERNAL
uidNumber:        DB uid
gidNumber:        DB gid
primary group:    FARM\<username>_gid
```

For a user assigned to a shared group:

```
AD user:          FARM\<username>
AD private group: FARM\<username>_gid
AD shared group:  FARM\<groupname>
principal:        <username>@FARM.DECS.INTERNAL
uidNumber:        DB uid
gidNumber:        shared group gid
primary group:    FARM\<groupname>
```

Important: AD `primaryGroupID` matters. If it remains `513` (`Domain Users`), new files can be created as `users` / `Domain Users` instead of the intended group.

Example commands:

```
sudo samba-tool group add ASCP || true
sudo samba-tool group addunixattrs ASCP 10070 || true

sudo samba-tool user add zio_ascp '<initial-password>' || true
sudo samba-tool user addunixattrs zio_ascp 10073 10070 || true
sudo samba-tool group addmembers ASCP zio_ascp || true
sudo samba-tool user setprimarygroup zio_ascp ASCP
```

Verify:

```
sudo samba-tool user show zio_ascp | \
  grep -E '^(sAMAccountName|objectSid|uidNumber|gidNumber|primaryGroupID):'

sudo samba-tool group show ASCP | \
  grep -E '^(sAMAccountName|objectSid|gidNumber):'
```

3. Synology NAS Setup

3.1 Join NAS To FARM AD

NAS must be joined to the FARM AD domain. This can be done from DSM UI or CLI. The CLI command used during testing was:

```
read -rsp "AD Administrator password: " AD_PASS; echo
sudo /usr/syno/sbin/synowin -joinDomain FARM Administrator "$AD_PASS" \
  -d 100.100.100.102 \
  -i 100.100.100.102 \
  -n FARM \
  -f farm.decs.internal
unset AD_PASS
```

Basic check:

```
/usr/syno/sbin/synowin -getWorkgroup
# Workgroup:FARM
```

Do not rely only on `synowin -updateDomain`; in our testing it could report `test join fail` even when enough AD/NFS pieces worked. The real checks are `id`, `wbinfo`, NFS write tests, and numeric `stat`.

3.1.1 NAS AD DC List

NAS must not depend on farm2 only. Keep farm2, farm6, and farm7 in the DNS, Kerberos, and Samba domain-member paths:

```
farm2 100.100.100.102
farm6 100.100.100.106
farm7 100.100.100.107
```

Current NAS files that carry this setting:

```
/etc/resolv.conf
/etc/resolv.conf.static
/etc/krb5.conf
/etc/samba/smb.conf
```

Expected DNS resolver configuration:

```
nameserver 100.100.100.102
nameserver 100.100.100.106
nameserver 100.100.100.107
```

Expected `/etc/krb5.conf` KDC list:

```
[realms]
  FARM.DECS.INTERNAL = {
    kdc = 100.100.100.102
    kdc = 100.100.100.106
    kdc = 100.100.100.107
    admin_server = 100.100.100.102
    kpasswd_server = 100.100.100.102:464
    kpasswd_server = 100.100.100.106:464
    kpasswd_server = 100.100.100.107:464
  }
```

Expected Samba domain member setting:

```
password server = 100.100.100.102 100.100.100.106 100.100.100.107
```

Apply changes with a backup first. The 2026-06-28 change used:

```
/root/decs-backups/nas-ad-multidc-20260628_204020
```

After changing these files:

```
SMB=/usr/local/packages/@appstore/SMBService/usr/bin
sudo "$SMB/net" cache flush || true
sudo systemctl restart pkg-synosamba-winbindd.service
sudo systemctl restart pkg-synosamba-smbd.service pkg-synosamba-nmbd.service || true
sudo "$SMB/net" cache flush || true
```

Verify:

```
nslookup -type=SRV _ldap._tcp.farm.decs.internal
sudo /usr/local/packages/@appstore/SMBService/usr/bin/wbinfo --online-status
sudo /usr/local/packages/@appstore/SMBService/usr/bin/wbinfo --dsgetdcname=FARM
sudo /usr/local/packages/@appstore/SMBService/usr/bin/wbinfo -i 'FARM\jy'
sudo /usr/local/packages/@appstore/SMBService/usr/bin/net ads testjoin
```

Important: NAS may still show a single `LDAP_server` / `KDC_server` in `/usr/syno/etc/private/adsinfo`. Treat that file as Synology's last observed domain info, not the complete failover list. The operational failover list is in resolver, Kerberos, and Samba configuration above.

3.2 NAS Samba / Winbind RFC2307

Back up first:

```
ts=$(date +%Y%m%d_%H%M%S)
sudo cp -a /etc/samba/smb.conf /etc/samba/smb.conf.decs-rfc2307-bak-$ts
sudo cp -a /etc/nsswitch.conf /etc/nsswitch.conf.decs-rfc2307-bak-$ts
```

Required `/etc/samba/smb.conf` global settings:

```
workgroup = FARM
realm = FARM.DECS.INTERNAL
security = ads
winbind:syno allow domains = FARM

idmap config FARM : backend = ad
idmap config FARM : schema_mode = rfc2307
idmap config FARM : range = 1000-299999
winbind nss info = rfc2307
winbind use default domain = yes
```

The `idmap config FARM : range` must include every DECS UID/GID used by DB and AD. The current FARM policy uses `1000-299999` because some valid identities are below `10000`.

Required `/etc/nsswitch.conf` shape on NAS:

```
passwd:    files winbind syno [NOTFOUND=return TRYAGAIN=return]
shadow:    files winbind syno [NOTFOUND=return TRYAGAIN=return]
group:     files winbind syno [NOTFOUND=return TRYAGAIN=return]
initgroups: files winbind syno [NOTFOUND=return TRYAGAIN=return]
```

Refresh:

```
SMB=/usr/local/packages/@appstore/SMBService/usr/bin
sudo "$SMB/net" cache flush || true
sudo systemctl restart pkg-synosamba-winbindd.service
```

Verify:

```
id 'FARM\hyrn2'
/usr/local/packages/@appstore/SMBService/usr/bin/wbinfo -i 'FARM\hyrn2'
```

Expected style:

```
uid=10069(FARM\hyrn2) gid=10070(FARM\ascp) groups=10070(FARM\ascp)
FARM\hyrn2:*:10069:10070:...
```

If NAS shows `96470xxx`, it is still using internal idmap for that identity. Do not run symbolic `chown -R 'FARM\user': 'FARM\group' ...` while that is true.

3.3 NAS Kerberos NFS Export

Current export policy for `/volume1/share`:

FARM storage IPs:

```
100.100.100.101 sec=krb5:krb5i:krb5p root_squash
100.100.100.102 sec=krb5:krb5i:krb5p root_squash
100.100.100.106 sec=krb5:krb5i:krb5p root_squash
100.100.100.107 sec=krb5:krb5i:krb5p root_squash
100.100.100.108 sec=krb5:krb5i:krb5p root_squash
100.100.100.109 sec=krb5:krb5i:krb5p root_squash
```

Management network:

```
192.168.2.11-19 sec=krb5:krb5i:krb5p no_root_squash
192.168.2.199 sec=sys:krb5p no_root_squash
DSM metadata/recovery-script fallback may still include 192.168.2.0/24 sec=sys no_root_squash
```

Core export line:

```
/volume1/share \
```

```
100.100.100.101(rw,async,no_wdelay,crossmnt,insecure,root_squash,insecure_locks,sec=krb5:krb5i:krb5p,anonuid=1025,anongid=100) \
```

```
100.100.100.102(rw,async,no_wdelay,crossmnt,insecure,root_squash,insecure_locks,sec=krb5:krb5i:krb5p,anonuid=1025,anongid=100) \
```

```
100.100.100.106(rw,async,no_wdelay,crossmnt,insecure,root_squash,insecure_locks,sec=krb5:krb5i:krb5p,anonuid=1025,anongid=100) \
```

```
100.100.100.107(rw,async,no_wdelay,crossmnt,insecure,root_squash,insecure_locks,sec=krb5:krb5i:krb5p,anonuid=1025,anongid=100) \
```

```
100.100.100.108(rw,async,no_wdelay,crossmnt,insecure,root_squash,insecure_locks,sec=krb5:krb5i:krb5p,anonuid=1025,anongid=100) \
```

```
100.100.100.109(rw,async,no_wdelay,crossmnt,insecure,root_squash,insecure_locks,sec=krb5:krb5i:krb5p,anonuid=1025,anongid=100) \
```

```
192.168.2.0/24(rw,async,no_wdelay,crossmnt,insecure,no_root_squash,insecure_locks,sec=sys,anonuid=1025,anongid=100)
```

Apply and refresh:

```
sudo exportfs -ra
```

```
now=$(date +%s)
```

```
for p in \
```

```
  /proc/net/rpc/auth.unix.gid/flush \
```

```
  /proc/net/rpc/nfs4.idtoname/flush \
```

```
  /proc/net/rpc/nfs4.nametoid/flush \
```

```
  /proc/net/rpc/auth.rpcsec.init/flush \
```

```
  /proc/net/rpc/auth.rpcsec.context/flush
```

```
do
```

```
  [ -w "$p" ] && printf "%s\n" "$now" | sudo tee "$p" >/dev/null || true
```

```
done
```

NAS GSS services:

```

sudo klist -kte /etc/nfs/krb5.keytab | grep -F 'nfs/nas.farm.decs.internal@FARM.DECS.INTERNAL'
sudo klist -kte /etc/nfs/krb5.keytab | grep -F 'nfs/nas.ai lab.dgu@AILAB.DGU'
sudo pkill svcgssd 2>/dev/null || true
sudo env KRB5_KTNAME=FILE:/etc/nfs/krb5.keytab /usr/sbin/svcgssd

sudo pkill idmapd 2>/dev/null || true
sudo /usr/sbin/idmapd

```

Verify:

```

sudo exportfs -v | sed -n '/\/volume1\/share/,+8p'
ps -ef | grep -E 'svcgssd|idmapd' | grep -v grep

```

NAS must have a valid service principal/keytab for:

```

nfs/nas.farm.decs.internal@FARM.DECS.INTERNAL
nfs/nas.ai lab.dgu@AILAB.DGU

```

The NAS serves both Kerberos realms through one `svcgssd`. Do not start it with `-p` `nfs/nas.farm.decs.internal@FARM.DECS.INTERNAL`; that constrains the acceptor to the FARM principal and prevents the AILAB service principal from being selected. With no `-p`, `svcgssd` acquires the host-based `nfs` service credential and MIT GSSAPI can select matching `nfs/<hostname>@<realm>` keys from the keytab. Do not substitute `svcgssd -n`: that requests a nameless acceptor and is broader than the required `nfs/*` service scope.

3.4 NAS Ownership Repair

Filesystem ownership is numeric. This command:

```

sudo chown -R 'FARM\hyrn2':'FARM\ASCP' /volume1/share/user-share/hyrn2

```

stores the numeric IDs resolved at that moment. If NAS was still resolving `FARM\hyrn2` to `96470xxx`, the wrong number was stored permanently until repaired.

After RFC2307 is verified, repair with numeric DB/AD IDs:

```
sudo chown -R 10073:10070 /volume1/share/user-share/zio_ascp
sudo chown -R 10071:10070 /volume1/share/user-share/ohchanju3
sudo chown -R 10011:10011 /volume1/share/user-share/zio1
sudo chown -R 10077:10078 /volume1/share/user-share/donghyun2
sudo chown -R 10102:10103 /volume1/share/user-share/suhyeon56
sudo chown -R 10069:10070 /volume1/share/user-share/hyrn2
sudo chown -R 10068:10068 /volume1/share/user-share/deepgu
```

Verify both views:

```
sudo stat -c '%U:%G %u:%g %A %n' /volume1/share/user-share/hyrn2
# FARM\hyrn2:FARM\ascp 10069:10070 drwxr-x---
```

4. FARM Host Setup

Every FARM host that mounts NAS NFS must be AD/RFC2307 aware. The container's local `/etc/passwd` is not enough because the NFS client is the host kernel.

4.1 Packages

```
sudo apt-get update
sudo apt-get install -y krb5-user nfs-common samba winbind libnss-winbind libpam-winbind
```

4.2 Kerberos Client Config

Install `/etc/krb5.conf` on every FARM host:

```
[libdefaults]
    default_realm = FARM.DECS.INTERNAL
    dns_lookup_realm = false
    dns_lookup_kdc = false
    rdns = false
    ticket_lifetime = 24h
    renew_lifetime = 7d
    forwardable = true

[realms]
    FARM.DECS.INTERNAL = {
        kdc = 100.100.100.102
        kdc = 100.100.100.106
        kdc = 100.100.100.107
        admin_server = 100.100.100.102
        default_domain = farm.decs.internal
    }

[domain_realm]
    .farm.decs.internal = FARM.DECS.INTERNAL
    farm.decs.internal = FARM.DECS.INTERNAL
```

4.3 Machine Keytab

Every host needs a root-only machine keytab for rpc.gssd:

```
/etc/krb5.keytab
```

Expected principals:

```
FARM1$@FARM.DECS.INTERNAL
FARM2$@FARM.DECS.INTERNAL
FARM6$@FARM.DECS.INTERNAL
FARM7$@FARM.DECS.INTERNAL
FARM8$@FARM.DECS.INTERNAL
FARM9$@FARM.DECS.INTERNAL
```

Verify:

```
short=$(hostname -s | tr '[:lower:]' '[:upper:]')
sudo kinit -k -t /etc/krb5.keytab "${short}@FARM.DECS.INTERNAL"
klist
kdestroy
```

Start rpc.gssd:

```
sudo systemctl enable --now rpc-gssd
sudo systemctl restart rpc-gssd
```

Do not override `rpc.gssd` with `-n`. Root performs the Kerberos NFS mount, so rpc.gssd must use the host machine keytab. The `-n` option makes root use user credentials instead, and caused FARM clients to lose the NAS mount when no matching root user ticket was available.

Check for and remove the bad override:

```
sudo systemctl cat rpc-gssd
sudo rm -f /etc/systemd/system/rpc-gssd.service.d/override.conf
sudo systemctl daemon-reload
sudo systemctl restart rpc-gssd
pgrep -a rpc.gssd
```

The expected process is `/usr/sbin/rpc.gssd` without `-n`.

4.4 Host Samba / NSS RFC2307

Recommended `/etc/samba/smb.conf` on domain-member FARM hosts:

```
[global]
workgroup = FARM
realm = FARM.DECS.INTERNAL
security = ADS
kerberos method = secrets and keytab

winbind enum users = yes
winbind enum groups = yes
winbind nss info = rfc2307
winbind use default domain = yes

idmap config * : backend = tdb
idmap config * : range = 3000000-3999999

idmap config FARM : backend = ad
idmap config FARM : schema_mode = rfc2307
idmap config FARM : range = 1000-299999
```

Join as domain member where applicable:

```
sudo net ads join -U Administrator
sudo systemctl restart smbd nmbd winbind
sudo systemctl enable winbind
```

Recommended `/etc/nsswitch.conf`:

```
passwd:      files systemd winbind
group:       files systemd winbind
shadow:      files
```

Verify:

```
getent passwd 'FARM\hyrn2'
getent group 'FARM\ASCP'
```

Expected style:

```
FARM\hyrn2:*:10069:10070:...
FARM\ASCP:x:10070:
```

If a host is not yet a clean AD member, a local static fallback can be used for recovery, but it should not be the long-term preferred state.

4.5 NFS Idmap Cache Flush

After changing identity mapping:

```
sudo nfsidmap -c 2>/dev/null || true
sudo sh -c 'ts=$(date +%s); for f in \
  /proc/net/rpc/auth.unix.gid/flush \
  /proc/net/rpc/nfs4.idtoname/flush \
  /proc/net/rpc/nfs4.nametoid/flush \
  /proc/net/rpc/auth.rpcsec.init/flush \
  /proc/net/rpc/auth.rpcsec.context/flush; do
  [ -w "$f" ] && echo "$ts" > "$f" || true
done'
```

5. FARM NFS Mount Setup

The active production mount is NFSv4.0 with Kerberos authentication:

```
source: nas.farm.decs.internal:/volume1/share
type:   nfs4
opts:
vers=4.0,rsiz=1048576,wsiz=1048576,sec=krb5,proto=tcp,hard,timeo=600,retrans=2,addr=100.100.100.120,_netdev,ex
xec,nouser
```

NFSv4.1 was tested and timed out or returned access denied during this rollout. Use NFSv4.0 until the NAS-side v4.1/RPCSEC_GSS behavior is separately fixed.

2026-07-06 large-write testing against the Synology NAS made this the current baseline profile. With `sec=krb5p,timeo=600,retrans=2`, FARM6/FARM8 and an earlier FARM7 test completed 50 GiB writes, but FARM1/FARM2/FARM9 reproduced D-state waits in the NFS `fsync` path. FARM2/FARM9 logged `nfs: server nas.farm.decs.internal not responding, still trying` and a stack through `nfs_file_fsync`. After changing the NAS export to allow `sec=krb5:krb5i:krb5p` and remounting clients with `sec=krb5i`, FARM1/2/6/7/8/9 all passed 50 GiB `dd ... conv=fsync` plus SHA256 verification. The current runtime was then lowered to `sec=krb5` for the internal FARM/NAS network by operator preference; that keeps Kerberos authentication and removes both integrity and privacy wrapping overhead. Keep the mount `hard` for production home/data paths; changing to `soft` can surface I/O errors to applications but risks partial writes or data corruption. Synology may still negotiate the effective runtime transfer size down to `rsiz=131072,wsiz=131072` even when the fstab requests 1 MiB. Treat the requested 1 MiB values as harmless desired values, and verify the actual runtime values with `findmnt` or `nfsstat -m`.

`server-state` owns the desired fstab entry. The monitoring-owned `decs-nfs-gss-mount-recovery.timer` recovers a missing mount only after fstab, DNS/storage RPC, Kerberos readiness, and D-state guards pass. The equivalent manual mount shape below is for troubleshooting and review, not the normal recovery entry point.

Manual mount shape for farm2:

```
sudo mkdir -p /home/tako2/share
sudo umount -f /home/tako2/share 2>/dev/null || sudo umount -l /home/tako2/share 2>/dev/null || true
sudo nfsidmap -c 2>/dev/null || true
sudo systemctl restart rpc-gssd || sudo systemctl start rpc-gssd
sudo mount -t nfs4 \
-o
vers=4.0,rsz=1048576,wsz=1048576,sec=krb5,proto=tcp,hard,timeo=600,retrans=2,addr=100.100.100.120,_netdev \
nas.farm.decs.internal:/volume1/share \
/home/tako2/share
```

`/etc/fstab` line shape:

```
nas.farm.decs.internal:/volume1/share /home/tako2/share nfs4
defaults,vers=4.0,rsz=1048576,wsz=1048576,sec=krb5,proto=tcp,hard,timeo=600,retrans=2,addr=100.100.100.120,
_netdev,exec,nouser 0 0
```

For each host, replace `/home/tako2/share` with the matching `/home/takoX/share`.

Verify:

```
findmnt -T /home/tako2/share -o TARGET,SOURCE,FSTYPE,OPTIONS
nfsstat -m | sed -n '/\/home\/tako2\/share/,+4p'
```

Expected:

```
nas.farm.decs.internal:/volume1/share
nfs4
vers=4.0
sec=krb5
addr=100.100.100.120
clientaddr=100.100.100.10X
```

It is expected that unprivileged host users such as `takoX` may see `Permission denied` on the Kerberos NFS mount if they do not have a Kerberos ticket. Verify mounts with `sudo findmnt` and verify user access through the container user's Kerberos ticket.

6. User Keytab And Ticket Cache

Per-user keytabs are root-only host secrets:

```
/etc/decs-krb/keytabs/<username>.keytab  
owner: root:root  
mode: 0400
```

The container must not receive the keytab. The host creates a user ticket cache:

```
/run/user/<uid>/krb5cc
```

Refresh env:

```
/etc/decs-krb/refresh.d/<username>.env
```

Required content:

```
DECS_KRB_PRINCIPAL=<username>@FARM.DECS.INTERNAL  
DECS_KRB_KEYTAB=/etc/decs-krb/keytabs/<username>.keytab  
DECS_KRB_CCACHE=FILE:/run/user/<uid>/krb5cc  
DECS_KRB_CCACHE_DIR=/run/user/<uid>  
DECS_KRB_UID=<uid>  
DECS_KRB_GID=<gid>
```

Refresh:

```
sudo rm -f /run/user/<uid>/krb5cc  
sudo systemctl start decs-krb-refresh@<username>.service  
sudo KRB5CCNAME=FILE:/run/user/<uid>/krb5cc klist
```

Expected:

```
Default principal: <username>@FARM.DECS.INTERNAL
```

Every FARM host that runs Kerberos containers must have the refresh helper and systemd template installed:

```
/usr/local/sbin/decs-krb-refresh  
/etc/systemd/system/decs-krb-refresh@.service  
/etc/systemd/system/decs-krb-refresh@.timer
```

Every active Kerberos user on that host must have one root-only env file and an enabled timer instance:

```
/etc/decs-krb/refresh.d/<username>.env  
decs-krb-refresh@<username>.timer
```

Operational check:

```
sudo find /etc/decs-krb/refresh.d -maxdepth 1 -type f -name '*.env' -printf '%f\n' |  
sort |  
while read f; do  
    u=${f%.env}  
    sudo systemctl enable --now "decs-krb-refresh@$u.timer"  
    systemctl is-active "decs-krb-refresh@$u.timer"  
    systemctl is-enabled "decs-krb-refresh@$u.timer"  
    sudo systemctl start "decs-krb-refresh@$u.service"  
done
```

Ticket check:

```
sudo bash -lc '  
for envfile in /etc/decs-krb/refresh.d/*.env; do  
    [ -e "$envfile" ] || continue  
    unset DECS_KRB_PRINCIPAL DECS_KRB_CCACHE  
    source "$envfile"  
    echo "== $(basename "$envfile" .env): $DECS_KRB_PRINCIPAL =="  
    KRB5CCNAME="$DECS_KRB_CCACHE" klist  
done'
```

After AD group changes, force a fresh keytab login at least once. A renewed ticket can keep old PAC/group information.

7. create-container Responsibilities

In FARM Kerberos mode, `create-container` must do all of this before recording a successful container:

1. Resolve or allocate DB UID/GID.
2. Ensure AD user `FARM\<username>` .

3. Ensure AD private group `FARM\<username>_gid`.
4. If `--group` differs from username, ensure shared group `FARM\<groupname>`.
5. Set AD user `uidNumber=<uid>` and `gidNumber=<gid>`.
6. Set AD target group `gidNumber=<gid>`.
7. Add user to target group and set AD primary group to the target group.
8. Export user keytab to target host as root-only secret.
9. Create or refresh `/run/user/<uid>/krb5cc`.
10. Ensure NAS home exists at `/volume1/share/user-share/<username>`.
11. Repair NAS home ownership numerically to `<uid>:<gid>`.
12. Flush NAS and host NFS/RPC identity caches.
13. Verify host-side NFS write as that Kerberos identity.
14. Run Docker with matching UID/GID and ccache bind mount.
15. Verify container `/home/<username>` ownership and read/write/delete.

The DECS image does not configure AD, NAS, host winbind, host keytabs, or NAS ownership. The image only consumes runtime values:

```
USER_ID=<username>
USER_GROUP=<groupname>
UID=<uid>
GID=<gid>
KRB5CCNAME=FILE:/run/user/<uid>/krb5cc
DECS_KRB5_PRINCIPAL=<username>@FARM.DECS.INTERNAL
DECS_KERBEROS_ENABLED=true
DECS_USER_SUDO_MODE=restricted
```

Docker run mount shape:

```
--mount type=bind,source=/home/takoX/share/user-share/,target=/home
--mount type=bind,source=/run/user/<uid>,target=/run/user/<uid>
--mount type=bind,source=/etc/krb5.conf,target=/etc/krb5.conf,readonly
```

8. User Home Group Sharing

Current working model:

Admins assign the user's runtime primary group at container creation time.
Users choose directories inside their own home.

If the user's primary group is the intended sharing group:

```
mkdir ~/sharing_dir
chmod 2770 ~/sharing_dir
```

Meaning:

```
2    setgid bit; new files/directories inherit the same group
770  owner and group have rwx, others have no access
```

During this rollout, normal-user `chgrp <group> <dir>` was rejected by Synology/NFS. Therefore the reliable current workflow is to set the intended shared group as the user's runtime primary group before the user creates shared directories.

9. Verification Matrix

9.1 AD

```
ssh -p 8082 jy@192.168.2.12 \
  "sudo samba-tool user show hyrn2 | grep -E '^(sAMAccountName|uidNumber|gidNumber|primaryGroupID|objectSid):'; \
  \
  sudo samba-tool group show ASCP | grep -E '^(sAMAccountName|gidNumber|objectSid):'"
```

9.2 NAS

```
ssh -p 6954 jy@192.168.2.30 \
  "id 'FARM\\hyrn2'; \
  /usr/local/packages/@appstore/SMBService/usr/bin/wbinfo -i 'FARM\\hyrn2'; \
  sudo stat -c '%U:%G %u:%g %A %n' /volume1/share/user-share/hyrn2"
```

9.3 FARM Host

```
ssh -p 8089 jy@192.168.2.19 \
  "findmnt -T /home/tako9/share -o TARGET,SOURCE,FSTYPE,OPTIONS; \
  getent passwd 'FARM\\hyrn2' || true; \
  getent group 'FARM\\ASCP' || true; \
  sudo stat -c '%U:%G %u:%g %A %n' /home/tako9/share/user-share/hyrn2"
```

9.4 Container

```

docker exec -u hyrn2 hyrn2_by_hr bash -lc '
set -euo pipefail
id
klist | sed -n "1,5p"
stat -c "%U:%G %u:%g %A %n" "$HOME"
base="$HOME/.decs_rw_verify_$(date +%s)_$$"
mkdir "$base"
printf one > "$base/a.txt"
printf two >> "$base/a.txt"
cp "$base/a.txt" "$base/b.txt"
mv "$base/b.txt" "$base/c.txt"
chmod 640 "$base/c.txt"
stat -c "DIR:%U:%G:%u:%g:%a:%n" "$base"
stat -c "FILE:%U:%G:%u:%g:%a:%n" "$base/a.txt"
stat -c "FILE:%U:%G:%u:%g:%a:%n" "$base/c.txt"
test "$(cat "$base/a.txt")" = onetwo
rm "$base/a.txt" "$base/c.txt"
rmdir "$base"
test ! -e "$base"
echo RESULT:ok
'
```

10. Current Verified Containers

The following containers passed UID/GID, ownership, create/read/append/copy, rename, chmod, stat, delete, and cleanup checks:

FARM1	zio_ascp	zio_ascp_by_hr	10073:10070	zio_ascp:ASCP	OK
FARM1	ohchanju3	ohchanju3_by_hr	10071:10070	ohchanju3:ASCP	OK
FARM2	zio1	zio2_jy	10011:10011	zio1:zio1	OK
FARM2	ohchanju3	ohchanju3_by_hr	10071:10070	ohchanju3:ASCP	OK
FARM6	donghyun2	donghyun2_by_sh	10077:10078	donghyun2:dscap26-1-3	OK
FARM8	suhyeon56	suhyeon56_by_sh	10102:10103	suhyeon56:headfirst	OK
FARM9	hyrn2	hyrn2_by_hr	10069:10070	hyrn2:ASCP	OK

deepgy_byhr was deleted and DB-soft-deleted:

```
server=FARM9
container=deepgy_byhr
existing=0
deleted_at=2026-06-27 19:45:54
```

11. Troubleshooting

Files show **nobody:nogroup**

Likely causes:

1. NAS is not using RFC2307 for FARM.
2. FARM host cannot resolve AD owner/group through NSS/winbind.
3. Old files still have numeric internal IDs from before RFC2307 was fixed.
4. Kerberos ticket is stale and still has old group/PAC data.

Checks:

```
# NAS
id 'FARM\<username>'
sudo stat -c '%U:%G %u:%g %n' /volume1/share/user-share/<username>

# FARM host
getent passwd 'FARM\<username>'
getent group 'FARM\<groupname>'
sudo stat -c '%U:%G %u:%g %n' /home/takoX/share/user-share/<username>

# Container
id
stat -c '%U:%G %u:%g %n' "$HOME"
```

New files are created as **users**

Check AD primary group:

```
sudo samba-tool user show <username> | grep -E '^(gidNumber|primaryGroupID):'
```

If **primaryGroupID** still points to Domain Users (**513**), set the primary group:

```
sudo samba-tool group addmembers <groupname> <username> || true
sudo samba-tool user setprimarygroup <username> <groupname>
sudo rm -f /run/user/<uid>/krb5cc
sudo systemctl start decs-krb-refresh@<username>.service
```

NFS mount uses 192.168.2.30

Remove it. That path is not the production Kerberos NFS transport.

```
findmnt -rn -t nfs,nfs4 | awk '$2 ~ /^192\.168\.2\.30:/ {print}'
```

The expected source is:

```
nas.farm.decs.internal:/volume1/share
addr=100.100.100.120
```

Monitor/exporter mount recovery uses 100.100.100.120:/volume1/share

Kerberos NFS recovery must also use the FQDN source. If a monitor or recovery service tries to mount 100.100.100.120:/volume1/share directly, Kerberos may look for an IP-based NFS service principal instead of:

```
nfs/nas.farm.decs.internal@FARM.DECS.INTERNAL
```

Required exporter defaults:

```
/etc/default/cluster-monitor-exporter:
CLUSTER_MONITOR_EXPORTER_REQUIRED_MOUNTS="nas.farm.decs.internal:/volume1/share=/home/takoX/share"

/etc/default/remote-boot-exporter:
REMOTE_BOOT_EXPORTER_REQUIRED_MOUNTS="nas.farm.decs.internal:/volume1/share=/home/takoX/share"
```

Replace takoX with the host number. The mount options can still pin the transport IP with addr=100.100.100.120 ; only the NFS source name must remain nas.farm.decs.internal .

rpc.gssd must not use -n

FARM root mounts rely on the host machine keytab in /etc/krb5.keytab . Do not create a systemd override like this:

```
[Service]
ExecStart=
ExecStart=/usr/sbin/rpc.gssd -n
```

That mode tells rpc.gssd to use user credentials for UID 0 instead of the machine keytab. In the 2026-07-03 incident, farm1/2/6/7/8 had this override and failed to mount `nas.farm.decs.internal:/volume1/share`; farm9 had no override and recovered first.

NAS NFS keytab KVNO health check and manual repair

NAS NFS GSS uses `/etc/nfs/krb5.keytab`. Since 2026-07-11, the FARM NFS principals are owned by the dedicated AD service account `svc-nfs-farm`, not the Synology machine account `NAS$`:

```
nfs/nas.farm.decs.internal@FARM.DECS.INTERNAL
nfs/NAS@FARM.DECS.INTERNAL
```

The `NAS$` account keeps only `HOST/*` and `RestrictedKrbHost/*` SPNs for Synology domain membership. Never rotate its password from this pipeline. `svc-nfs-farm` has `msDS-SupportedEncryptionTypes=28` and does not expire automatically. Its current `userAccountControl=66048` includes `DONT_EXPIRE_PASSWORD`, and `accountExpires=0`. No rotation timer is installed. KVNO changes only if an administrator explicitly runs the manual `--rotate --apply` command below.

The monitoring checker connects to farm2 for the `svc-nfs-farm` KVNO and reads the NAS keytab metadata. It verifies the AILAB principal, validates current service-ticket decryption with `kvno -k`, and writes a local status JSON. It never changes AD, writes a keytab, restarts a service, or remounts NFS.

Repair is a separate explicit operation. It exports fresh keys, preserves unrelated AILAB principals, atomically replaces the NAS keytabs, and may restart `svcgssd` without `-p`:

```
/home/jy/server_manage/monitoring/health-checks/kerberos-nfs-keytab/script/check-nfs-keytab.sh --profile farm
/home/jy/server_manage/kerberos-nfs/script/keytab/repair-farm-nas-nfs-keytab.sh --repair
/home/jy/server_manage/kerberos-nfs/script/keytab/rotate-farm-nfs-service-key.sh --check
# Planned maintenance only:
/home/jy/server_manage/kerberos-nfs/script/keytab/rotate-farm-nfs-service-key.sh --rotate --apply
systemctl --user import-environment SSH_AUTH_SOCKET
cd /home/jy/server_manage/monitoring
ANSIBLE_CONFIG=ansible_playbook/ansible.cfg \
ansible-playbook ansible_playbook/deploy_kerberos_nfs_keytab_check.yml
systemctl --user list-timers 'check-nfs-keytab@farm.timer'
```

The checker timer runs every five minutes. Manual repair and key rotation are never invoked by that timer.

The script backs up replaced NAS keytabs as:

```
/etc/krb5.keytab.before-decs-watch-YYYYmdd_HHMMSS
/etc/nfs/krb5.keytab.before-decs-watch-YYYYmdd_HHMMSS
```

Keep the previous FARM kvno entries for at least 48 hours, exceeding the configured 24-hour ticket lifetime. Remove older entries only in a later maintenance operation after FARM and AILAB canaries are healthy. If a newly generated kvno number collides with a retained `NAS$` kvno, the watcher removes the conflicting retained FARM entries before appending the verified `svc-nfs-farm` keys; unrelated AILAB entries are never pruned.

NFSv4.1

NAS advertises NFSv4.1, but `vers=4.1,sec=krb5p` timed out or failed during the rollout. Production is currently `vers=4.0,sec=krb5` with `timeo=600,retrans=2`. Re-test v4.1 only after isolating the NAS RPCSEC_GSS behavior.

12. Useful Local Files

```
/home/jy/server_manage/monitoring/health-checks/kerberos-nfs-keytab/
/home/jy/server_manage/kerberos-nfs/script/keytab/repair-farm-nas-nfs-keytab.sh
/home/jy/server_manage/kerberos-nfs/script/keytab/rotate-farm-nfs-service-key.sh
/home/jy/server_manage/kerberos-nfs/docs/farm.md
/home/jy/server_manage/kerberos-nfs/docs/lab.md
/home/jy/server_manage/user-lifecycle/README.md
/home/jy/server_manage/user-lifecycle/README_create_container.md
```

13. Actual Extra / Residual Settings Observed

This section records settings that exist on the servers but are not part of the clean conceptual core. They are either intentional compatibility settings, temporary recovery aids, or unrelated existing mounts.

NAS

NAS currently has the expected RFC2307 settings:

```
idmap config FARM : backend = ad
idmap config FARM : schema_mode = rfc2307
idmap config FARM : range = 1000-299999
winbind nss info = rfc2307
```

Additional observed details:

1. /volume1/share still exports 192.168.2.0/24 with sec=sys,no_root_squash.
This is a management/backward-compatibility export. FARM production mounts must not use it.
2. /volume1/test_krb still exists as a Kerberos test export.
It is not used by current production user-share containers.
3. /etc/nsswitch.conf has duplicate initgroups lines.
This should not affect the current working state, but can be cleaned later.
4. A numeric ownership repair job was still running during this audit:
/tmp/decs_rfc2307_numeric_chown_.log

FARM hosts

Expected FARM user-share mounts are NFSv4.0 Kerberos mounts:

```
nas.farm.decs.internal:/volume1/share
vers=4.0
sec=krb5
hard
timeo=600
retrans=2
addr=100.100.100.120
```

If `findmnt` shows `sec=sys`, `sec=krb5p`, or `timeo=50,retrans=1`, the host is still using a legacy runtime superblock. Apply the desired `sec=krb5` fstab entry through `server-state`; monitoring will recover the mount only after the expected configuration and readiness checks pass. For an already mounted legacy superblock, cleanly unmount/remount during a maintenance window. If NFS callers are in D-state, do not repeat forced unmounts; reboot the host.

Additional observed details:

1. farm1, farm8, and farm9 still show Samba server role = standalone server.
They are not clean AD domain members yet.
2. Local static fallback entries for FARM\jy_gid exist in /etc/group and /etc/idmapd.conf on FARM hosts. These are recovery leftovers for the jy identity work, not the desired long-term model for all users.
3. farm2, farm6, and farm7 are Samba AD DC hosts.
4. farm6 has an unrelated existing fstab line:
`100.100.100.120:/volume1/bai_mnt /vfroot/shared nfs defaults,exec,nouser 0 0`

AD

AD user/group RFC2307 values for the repaired users are correct on farm2.

Observed remaining issue:

Samba AD replication between farm2/farm6/farm7 is still failing.
Until that is fixed, perform AD writes against farm2 and verify the target identity from the NAS/FARM host actually used by the container.

UID repository

The runtime servers are configured as above. Separately, the local UID repo has uncommitted development changes on branch `uid260627-krb-operational`; this is not a host runtime setting, but it matters before committing/merging the final create-container implementation.

Kerberos Ticket Refresh Margin

Updated on 2026-06-29.

DECS containers do not receive user keytabs. Each FARM Docker host keeps the user keytab as a root-only secret under `/etc/decs-krb/keytabs` and exposes only the host ccache at `/run/user/<uid>/krb5cc` to the container.

Ticket refresh is handled by:

```
/usr/local/sbin/decs-krb-refresh
/etc/systemd/system/decs-krb-refresh@.service
/etc/systemd/system/decs-krb-refresh@.timer
/etc/decs-krb/refresh.d/<username>.env
```

The timer normally runs every 1 hour. The refresh script first inspects the existing ccache. If the Kerberos `renew until` deadline is more than 24 hours away, it performs `kinit -R`. If `renew until` is within 24 hours, missing, or the renewal fails, it uses the root-only keytab to issue a fresh ticket:

```
kinit -k -t /etc/decs-krb/keytabs/<username>.keytab \  
-c FILE:/run/user/<uid>/krb5cc \  
<username>@FARM.DECS.INTERNAL
```

The 24-hour margin is controlled by `DECS_KRB_REISSUE_BEFORE_SECONDS` and defaults to `86400`. It can be overridden per user by adding the variable to `/etc/decs-krb/refresh.d/<username>.env`.

Operational checks:

```
systemctl list-timers 'decs-krb-refresh@*.timer'  
systemctl status decs-krb-refresh@<username>.timer  
systemctl start decs-krb-refresh@<username>.service  
klist -c FILE:/run/user/<uid>/krb5cc
```